

Scope Boundary Reference

Scope-boundary **diamond** symbols mark the **“last-meter” hand-over** between **in-scope** assets (designed/supplied within the MINERVA CryoCell package) and **out-of-scope** assets (infrastructure, building services, RF system delivered by others). They appear on **Layer 1 — Scope** of every sheet, with the hand-over note on **Layer 12 — Tags**.

1. Diamond code format — TPXYYYY

TP

X

YYYY

4-digit serial number (0001, 0002, ...)

category prefix (one letter, see table below)

"TP" = Terminal Point / scope interface

2. Category prefixes

Prefix	Category	Typical media / service
B	Bulk	Bulk gas / cryogen supply
C	Cryogenic	He circuits (2 K / 4.5 K / 40 K), guard
E	Electrical	Heater power, electrical feed-throughs
H	HVAC	Vacuum / venting / ventilation, relief
L	Liquid	Liquid level / liquid services
S	Steam	Steam / hot utilities
W	Water	DI cooling water, FREIA water

The category is assigned automatically from the **function of the nearest instrument** at each terminal point (e.g. a point next to an electric heater → E , next to a level switch → L , next to a relief/vacuum device → H).

3. In-scope vs out-of-scope convention

- **Inside** the diamond ring → in-scope (MINERVA CryoCell).
- **Outside** the diamond ring → out-of-scope (delivered/maintained by others).
- The diamond is the single contractual/physical interface — the “last meter” of pipe/cable on the in-scope side terminates at the diamond.

4. Terminal-point register (QCELL / LB)

Code	Cat	Category name	Source layer
TPC0001	C	Cryogenic	Terminal points
TPC0002	C	Cryogenic	Terminal points
TPH0003	H	HVAC	Terminal points
TPC0004	C	Cryogenic	Terminal points
TPC0005	C	Cryogenic	Terminal points
TPC0006	C	Cryogenic	Terminal points
TPE0007	E	Electrical	Terminal points
TPC0008	C	Cryogenic	Terminal points
TPC0009	C	Cryogenic	Terminal points
TPE0010	E	Electrical	Terminal points
TPE0011	E	Electrical	Terminal points
TPC0012	C	Cryogenic	Terminal points
TPC0013	C	Cryogenic	Terminal points
TPC0014	C	Cryogenic	Terminal points
TPC0015	C	Cryogenic	Terminal points
TPE0016	E	Electrical	Terminal points
TPE0017	E	Electrical	Terminal points
TPC0018	C	Cryogenic	Terminal points
TPC0019	C	Cryogenic	Terminal points
TPC0020	C	Cryogenic	Terminal points
TPL0021	L	Liquid	Terminal points
TPL0022	L	Liquid	Terminal points

RFCELL terminal points are not individually tagged in the source extract;
 RFCELL scope hand-over is represented by the same diamond convention where
 coupler/water interfaces leave the cell.

5. Machine-readable

The full code list (per drawing) is stored in `_build_meta.json` under `scope_codes` .